

Momentum

A general purpose blockchain platform utilizing proof-of-popularity

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Abstract—This document outlines the key concepts and operational mechanics of a proposed blockchain system utilizing a novel consensus algorithm called proof-of-popularity.

Index Terms—decentralized systems, consensus, blockchain, smart contracts, cryptocurrency

I. INTRODUCTION

Momentum is an implementation of a blockchain system utilizing the proof-of-popularity mechanism.

Most conventional blockchain systems rely on a consensus mechanism which is either based on proof-of-work or proof-of-stake. Both come with their own set of problems: Proof-of-work requires a large amount of computational power, which draws a lot of criticism for its wastefulness in our resource-constrained world. Proof-of-stake usually involves pre-mining, which assigns a large amount of its value to predetermined wallet addresses.

Proof-of-popularity addresses these problems and offers an alternative to the aforementioned consensus mechanism.

II. OVERVIEW

In a proof-of-popularity-based blockchain, there exists one block for each timestamp, while the valid chain is always the chain that contains the block with the highest popularity. Nodes work together to determine the popularity of any given block and ensure its integrity and thus the integrity of the network itself. To guarantee the finality of any given block, it must be prohibitively difficult to forge another block with the same timestamp, but higher popularity.

On a high level, a well-behaved node in the Momentum network acts like this:

- 1) Await total popularity advertisements from peers.
- 2) Download block with highest popularity
- 3) Validate block
- 4) Await next timestamp
- 5) Generate new block candidate
- 6) Send block hash to peers
- 7) Await block hashes and signatures
- 8) Send signature to the producer of the block with the highest *totalrank*.
- 9) Append block candidate to chain and advertise the new popularity
- 10) Go to 1

III. BLOCK VALIDATION

A block is valid if:

- There is no block with the same or lower timestamp which has a higher popularity.
- The referenced previous block is valid.
- The timestamp of the block is exactly 1 greater than the timestamp of the referenced previous block.
- The timestamp of the block is lower than or equal to the current timestamp.
- The *rank* value is correct
- The *totalrank* value equals the *totalrank* value of the referenced previous block plus the *rank* value of the block.

IV. DEFINITIONS

A. Node

A participating Machine (Computer) in the Momentum Network.

B. Rank

For each timestamp there exists a global order of all accounts that hold Momentum. This order is calculated using a randomized weighting function that takes a timestamp and the set of all accounts as input, and returns an ordered list of accounts. This order is used to determine the priority for each account to produce the next block. The rank of the block producers is included in the block header.

Each block has a *rank* and *totalrank* value within its header. The *rank* value specifies its rank in the priority list during the timestamp of block production. The *totalrank* value is the sum of all *rank* values in the chain.

C. Popularity

The popularity of a block. This is the sum of the popularity of the previous block plus the weight of all collected signatures included in the block header. The chain that contains the most popular block is the valid (authorative) chain.

V. IMPLEMENTATION

A. Storage

While it's up to the implementation to decide on its data storage strategy, it makes sense to utilize a distributed hash table and leverage already existing implementations.

B. Peer Discovery

TODO: Utilize DHT implementation, for example Kademlia

VI. CONSIDERATIONS

A. Network speed

The distribution of the last block might take a long time, especially as the network grows and the number of signatures per block increases. The timestamp interval has to be chosen in a way to allow for a good amount of time for the last block to propagate through the DHT to many nodes.

VII. CONCLUSION

Momentum and its proof-of-popularity-based approach offers many benefits over conventional blockchain systems. For one, it does not utilize proof-of-work, and as such fits nicely into a resource-constrained and climate-conscious environment. There also is no pre-mining of any form, which is usually required in a proof-of-stake blockchain.